SAECOIN — MAINNET LAUNCH CHECKLIST

Document: Mainnet_Launch_Checklist.md
Project: SAECOIN
Ticker: SAC (Provisional)
Version: 1.0
Status: Planning
Target Platform: Windows 11 (16 GB RAM development environment)
Purpose: Mainnet Readiness, Verification and Launch Approval

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1. PURPOSE

This document defines the technical, security, operational and organizational requirements that must be completed before launching the SAECOIN mainnet.

A completed checklist does not automatically authorize launch. Mainnet launch requires documented verification and formal approval.

All consensus-critical parameters must be finalized before implementation is considered production-ready.

Status Definitions

- [ ] Not Started
- [~] In Progress
- [x] Completed
- [!] Blocked
- [N/A] Not Applicable

Each completed item should have supporting evidence.

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2. PROTOCOL FINALIZATION

The blockchain protocol must be fully specified before mainnet launch.

2.1 Consensus Parameters

- [ ] Select and finalize the Proof of Work algorithm.
- [ ] Finalize the block hashing algorithm.
- [ ] Finalize the difficulty adjustment formula.
- [ ] Define target block time.
- [ ] Define maximum and minimum difficulty rules.
- [ ] Define timestamp validation rules.
- [ ] Define acceptable timestamp drift.

- [ ] Finalize block header serialization.
- [ ] Finalize block serialization.
- [ ] Finalize transaction serialization.
- [ ] Define chain selection rules.
- [ ] Define fork and reorganization rules.
- [ ] Define maximum block size or weight.
- [ ] Define maximum transaction size.
- [ ] Define coinbase maturity.
- [ ] Define transaction fee rules.
- [ ] Define mempool acceptance rules.
- [ ] Define invalid block rejection rules.

2.2 Monetary Policy

Proposed parameters requiring final verification:

Parameter| Proposed Value
Maximum Supply| 21,000,000 SAC
Initial Block Reward| 50 SAC
Target Block Time| 10 Minutes
Halving Interval| 210,000 Blocks
Premine| 0 SAC
Decimal Precision| 8

Checklist:

- [ ] Verify the complete mathematical issuance schedule.
- [ ] Confirm maximum supply under integer arithmetic.
- [ ] Define subsidy rounding rules.
- [ ] Define the final subsidy era.
- [ ] Define what happens when block subsidy reaches zero.
- [ ] Verify that no transaction can create unauthorized coins.
- [ ] Verify coinbase reward validation.
- [ ] Verify total supply accounting.
- [ ] Document all monetary rules.

Important: The proposed reward and halving schedule must not be assumed to produce exactly 21 million SAC without mathematical verification.

2.3 Network Parameters

- [ ] Finalize mainnet network identifier.
- [ ] Finalize message magic bytes.
- [ ] Finalize default P2P port.
- [ ] Finalize RPC configuration.
- [ ] Finalize address prefix.
- [ ] Finalize key and signature formats.
- [ ] Finalize transaction version rules.

- [ ] Finalize block version rules.
- [ ] Define protocol upgrade rules.
- [ ] Define minimum supported node version.
- [ ] Document network compatibility requirements.

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3. GENESIS BLOCK VERIFICATION

The genesis block is a permanent foundation of the blockchain.

- [ ] Finalize genesis block configuration.
- [ ] Finalize genesis timestamp.
- [ ] Finalize genesis message, if applicable.
- [ ] Finalize genesis coinbase transaction.
- [ ] Confirm zero premine, if retained.
- [ ] Generate genesis block using the official implementation.
- [ ] Independently verify genesis block hash.
- [ ] Verify genesis transaction hash.
- [ ] Verify genesis Merkle root.
- [ ] Verify genesis proof of work.
- [ ] Verify genesis configuration across all node builds.
- [ ] Confirm every node generates or recognizes the same genesis block.
- [ ] Publish the genesis block hash.
- [ ] Store the final genesis configuration and verification evidence.

NO-GO CONDITION: If different official nodes produce or recognize different genesis hashes, launch must be stopped.

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4. BLOCKCHAIN CORE VALIDATION

4.1 Transaction Validation

- [ ] Validate transaction inputs.
- [ ] Validate transaction outputs.
- [ ] Prevent double spending.
- [ ] Validate transaction signatures.
- [ ] Validate input and output amounts.
- [ ] Prevent integer overflow.
- [ ] Reject invalid transaction formats.
- [ ] Reject unauthorized coin creation.
- [ ] Validate transaction fees.
- [ ] Validate coinbase transactions.
- [ ] Test transaction ordering and dependency handling.

4.2 Block Validation

- [ ] Validate block headers.
- [ ] Validate proof of work.
- [ ] Validate difficulty.
- [ ] Validate timestamps.
- [ ] Validate Merkle roots.
- [ ] Validate block rewards.
- [ ] Validate all block transactions.
- [ ] Reject duplicate transactions where prohibited.
- [ ] Reject invalid blocks.
- [ ] Test chain reorganizations.
- [ ] Test competing valid chains.
- [ ] Verify deterministic validation across nodes.

4.3 UTXO Validation

- [ ] Verify UTXO creation.
- [ ] Verify UTXO spending.
- [ ] Prevent duplicate input spending.
- [ ] Verify spent-output handling.
- [ ] Verify database consistency.
- [ ] Verify UTXO recovery after restart.
- [ ] Test corrupted database recovery.
- [ ] Verify UTXO state after chain reorganization.

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5. MINING VALIDATION

5.1 CPU Mining

- [ ] Implement CPU mining.
- [ ] Verify valid block generation.
- [ ] Verify invalid block rejection.
- [ ] Test different CPU configurations.
- [ ] Measure CPU usage.
- [ ] Measure mining performance.
- [ ] Test long-duration mining.
- [ ] Verify mining does not corrupt node state.

5.2 GPU Mining

- [ ] Finalize GPU mining architecture.
- [ ] Implement GPU mining support.
- [ ] Test supported GPU hardware.
- [ ] Verify GPU-generated proofs.
- [ ] Compare CPU and GPU verification results.
- [ ] Test GPU error handling.

- [ ] Measure GPU performance.
- [ ] Test long-duration GPU mining.

5.3 Mining Economics and Fairness

- [ ] Verify mining reward calculations.
- [ ] Verify difficulty adjustment under changing hash rate.
- [ ] Test low hash-rate conditions.
- [ ] Test sudden hash-rate increases.
- [ ] Test mining after difficulty adjustments.
- [ ] Document solo mining requirements.
- [ ] Document future mining pool compatibility considerations.

Important: ASIC resistance is a design objective, not a guarantee. Hardware resistance must be evaluated through independent analysis and real-world testing.

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6. P2P NETWORK VALIDATION

- [ ] Verify peer discovery.
- [ ] Verify peer connection handling.
- [ ] Verify block propagation.
- [ ] Verify transaction propagation.
- [ ] Verify peer disconnection handling.
- [ ] Test nodes behind NAT.
- [ ] Test unstable network connections.
- [ ] Test network partitions.
- [ ] Test peer reconnection.
- [ ] Prevent malformed messages from crashing nodes.
- [ ] Apply connection limits.
- [ ] Apply message size limits.
- [ ] Apply rate limits where appropriate.
- [ ] Test malicious peer behavior.
- [ ] Verify synchronization from genesis.
- [ ] Verify synchronization from bootstrap peers.
- [ ] Test network recovery after temporary outages.

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7. WALLET SECURITY AND VALIDATION

7.1 Key Management

- [ ] Finalize wallet key generation standard.
- [ ] Finalize signature algorithm.
- [ ] Finalize seed phrase or recovery format.
- [ ] Use a cryptographically secure random number generator.

- [ ] Verify private key generation.
- [ ] Verify public key derivation.
- [ ] Verify address generation.
- [ ] Verify signature creation and verification.
- [ ] Prevent private keys from appearing in logs.
- [ ] Prevent private keys from being transmitted unnecessarily.
- [ ] Review memory handling of sensitive key material.

7.2 Wallet Functions

- [ ] Create wallet.
- [ ] Import wallet.
- [ ] Restore wallet.
- [ ] Display receiving address.
- [ ] Send SAC.
- [ ] Receive SAC.
- [ ] Display transaction history.
- [ ] Display confirmations.
- [ ] Display available and pending balances.
- [ ] Handle insufficient funds.
- [ ] Handle invalid addresses.
- [ ] Handle network interruptions.
- [ ] Verify transaction fee calculation.

7.3 Backup and Recovery

- [ ] Test wallet backup.
- [ ] Test wallet restoration on a different device.
- [ ] Verify restored addresses.
- [ ] Verify restored balances after synchronization.
- [ ] Test incorrect recovery phrase handling.
- [ ] Document safe backup practices.
- [ ] Warn users against sharing recovery phrases.
- [ ] Test encrypted wallet backup, if supported.

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8. SECURITY AUDIT

An independent security review should be completed before mainnet launch.

8.1 Code Security

- [ ] Review Rust code for memory and logic safety.
- [ ] Review cryptographic implementation.
- [ ] Review transaction validation.
- [ ] Review consensus validation.
- [ ] Review P2P message parsing.

- [ ] Review wallet key handling.
- [ ] Review database access.
- [ ] Review RPC security.
- [ ] Review dependency vulnerabilities.
- [ ] Remove unnecessary dependencies.
- [ ] Review permissions and secret handling.

8.2 Testing and Analysis

- [ ] Run unit tests.
- [ ] Run integration tests.
- [ ] Run end-to-end tests.
- [ ] Run fuzz testing.
- [ ] Run property-based tests.
- [ ] Run malformed transaction tests.
- [ ] Run malformed block tests.
- [ ] Run P2P abuse tests.
- [ ] Run integer overflow tests.
- [ ] Run denial-of-service resilience tests.
- [ ] Perform independent consensus review.
- [ ] Perform independent cryptographic review.
- [ ] Document all discovered vulnerabilities.
- [ ] Verify all critical and high-severity findings are resolved.

NO-GO CONDITION: Any unresolved critical or high-severity vulnerability blocks mainnet launch.

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9. TESTING AND PERFORMANCE

9.1 Functional Testing

- [ ] Test node installation.
- [ ] Test node startup.
- [ ] Test node shutdown.
- [ ] Test blockchain synchronization.
- [ ] Test block creation.
- [ ] Test transaction creation.
- [ ] Test transaction confirmation.
- [ ] Test wallet synchronization.
- [ ] Test wallet backup and restoration.
- [ ] Test explorer data accuracy.

9.2 Performance Testing

- [ ] Measure block validation speed.
- [ ] Measure transaction validation speed.

- [ ] Measure block propagation time.
- [ ] Measure synchronization speed.
- [ ] Measure memory consumption.
- [ ] Measure disk usage.
- [ ] Measure CPU usage.
- [ ] Measure GPU mining performance.
- [ ] Test long-running node stability.
- [ ] Test database growth.
- [ ] Test recovery after unexpected shutdown.

9.3 Windows Compatibility

Development environment:

- Windows 11
- 16 GB RAM

Checklist:

- [ ] Build node on the target Windows environment.
- [ ] Run node on Windows 11.
- [ ] Build wallet applications.
- [ ] Test wallet installation.
- [ ] Test wallet updates.
- [ ] Test node and wallet communication.
- [ ] Verify firewall requirements.
- [ ] Verify data directory permissions.
- [ ] Verify clean installation and uninstallation.

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10. PUBLIC TESTNET

A public testnet must operate before mainnet launch.

10.1 Testnet Deployment

- [ ] Finalize testnet network parameters.
- [ ] Generate a separate testnet genesis block.
- [ ] Deploy public seed nodes.
- [ ] Publish testnet node software.
- [ ] Publish testnet wallet.
- [ ] Publish testnet mining instructions.
- [ ] Publish testnet network information.
- [ ] Invite external testers.

10.2 Testnet Operations

- [ ] Verify node synchronization.
- [ ] Verify CPU mining.
- [ ] Verify GPU mining.
- [ ] Verify transaction processing.
- [ ] Verify wallet restoration.
- [ ] Test block reorganizations.
- [ ] Test network partitions.
- [ ] Test node crashes and recovery.
- [ ] Test database recovery.
- [ ] Test malicious peer behavior.
- [ ] Test difficulty adjustments.
- [ ] Test long-duration network operation.
- [ ] Fix and retest reported bugs.

10.3 Testnet Exit Criteria

- [ ] No unresolved critical consensus bugs.
- [ ] No unresolved critical security vulnerabilities.
- [ ] Nodes remain stable during extended operation.
- [ ] Independent nodes agree on the same chain.
- [ ] Monetary issuance matches the specification.
- [ ] Wallet backup and restoration work reliably.
- [ ] Mining and difficulty adjustment behave as specified.
- [ ] Recovery procedures have been tested.

The required testnet duration should be decided and documented before testing begins. A short successful demonstration is not sufficient evidence of production readiness.

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11. EXPLORER AND API READINESS

- [ ] Deploy blockchain explorer backend.
- [ ] Deploy explorer frontend.
- [ ] Verify block information.
- [ ] Verify transaction information.
- [ ] Verify address information.
- [ ] Verify supply calculations.
- [ ] Verify confirmation counts.
- [ ] Verify API responses.
- [ ] Apply API rate limits.
- [ ] Prevent unauthorized administrative access.
- [ ] Configure HTTPS.
- [ ] Configure monitoring.
- [ ] Test explorer recovery.
- [ ] Verify explorer data against independent nodes.

The explorer must not be treated as the authoritative source of consensus. Full nodes independently validate the blockchain.

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12. RELEASE PREPARATION

12.1 Official Software

- [ ] Prepare official full-node release.
- [ ] Prepare official wallet release.
- [ ] Prepare mining software release.
- [ ] Prepare explorer release.
- [ ] Prepare installation instructions.
- [ ] Prepare upgrade instructions.
- [ ] Prepare release notes.
- [ ] Prepare troubleshooting documentation.

12.2 Build Verification

- [ ] Build from a clean source checkout.
- [ ] Verify release build results.
- [ ] Verify dependencies and versions.
- [ ] Generate binary checksums.
- [ ] Sign release artifacts where supported.
- [ ] Verify release signatures.
- [ ] Test installation packages.
- [ ] Test software on clean systems.
- [ ] Preserve source code and release artifacts.
- [ ] Publish the official source tag.

12.3 Public Documentation

- [ ] Publish SAECOIN whitepaper or protocol overview.
- [ ] Publish PRD and technical documentation as appropriate.
- [ ] Publish network parameters.
- [ ] Publish genesis block hash.
- [ ] Publish wallet instructions.
- [ ] Publish node instructions.
- [ ] Publish mining instructions.
- [ ] Publish security contact information.
- [ ] Publish official project channels.
- [ ] Clearly identify provisional or unsupported features.

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13. MAINNET INFRASTRUCTURE

13.1 Nodes

- [ ] Deploy official mainnet seed nodes.
- [ ] Deploy geographically distributed nodes.
- [ ] Configure node monitoring.
- [ ] Configure firewall rules.
- [ ] Configure secure remote administration.
- [ ] Configure automated restart policies where appropriate.
- [ ] Test node recovery.
- [ ] Verify independent operators can run nodes.

13.2 Monitoring

Monitor at least:

- Block height.
- Block production interval.
- Network difficulty.
- Network hash rate estimates.
- Peer count.
- Block propagation.
- Node synchronization status.
- Disk usage.
- Memory usage.
- CPU usage.
- Wallet and API availability.
- Reorganization events.
- Unexpected chain divergence.

13.3 Backups

- [ ] Back up critical configuration.
- [ ] Back up release artifacts.
- [ ] Back up explorer data where required.
- [ ] Securely store signing keys.
- [ ] Restrict access to infrastructure secrets.
- [ ] Test restoration procedures.
- [ ] Document backup responsibilities.

Private wallet keys must never be stored in ordinary server backups without appropriate protection.

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14. GOVERNANCE AND INCIDENT RESPONSE

- [ ] Define protocol change procedures.
- [ ] Define software release approval procedures.

- [ ] Define emergency vulnerability response.
- [ ] Define security disclosure process.
- [ ] Define node operator communication channels.
- [ ] Define incident escalation procedures.
- [ ] Define network upgrade coordination.
- [ ] Document how consensus-critical changes are reviewed.
- [ ] Document how users are informed about security incidents.
- [ ] Establish a process for publishing post-incident reports.

No administrator should be able to secretly alter consensus rules or create unauthorized SAC.

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15. LEGAL AND REGULATORY REVIEW

Before any public sale, exchange listing, fundraising, or commercial launch:

- [ ] Obtain appropriate legal advice.
- [ ] Review applicable Malaysian laws and regulations.
- [ ] Review regulations in every intended operating jurisdiction.
- [ ] Review digital asset and payment-related requirements.
- [ ] Review consumer protection obligations.
- [ ] Review privacy and data protection obligations.
- [ ] Review advertising and financial promotion restrictions.
- [ ] Review exchange listing requirements before approaching exchanges.
- [ ] Prepare appropriate terms and risk disclosures.

This checklist does not establish legal compliance. Regulatory obligations depend on the project's actual activities and jurisdictions.

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16. FINAL MAINNET CONFIGURATION

Before launch, create a version-controlled and independently reviewed mainnet configuration manifest.

It should include:

- Network name.
- Network identifier.
- Network magic bytes.
- P2P port.
- RPC configuration.
- Consensus algorithm.
- Hash algorithm.
- Difficulty adjustment parameters.

- Block timing parameters.
- Block size or weight limits.
- Transaction limits.
- Monetary issuance parameters.
- Address format.
- Signature format.
- Genesis block configuration.
- Genesis block hash.
- Protocol version.
- Software release version.
- Official bootstrap peer information.

Checklist:

- [ ] Configuration reviewed.
- [ ] Configuration independently verified.
- [ ] Genesis hash verified.
- [ ] Issuance schedule verified.
- [ ] Mainnet and testnet configurations separated.
- [ ] Configuration committed to the official repository.
- [ ] Final configuration checksum recorded.

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17. LAUNCH DAY CHECKLIST

Before Launch

- [ ] Formal launch approval received.
- [ ] Mainnet configuration frozen.
- [ ] Genesis block independently verified.
- [ ] Official node binaries verified.
- [ ] Official wallet binaries verified.
- [ ] Release checksums published.
- [ ] Seed nodes online.
- [ ] Explorer online.
- [ ] Monitoring active.
- [ ] Support channels ready.
- [ ] Incident response team available.
- [ ] Public launch announcement prepared.

At Launch

- [ ] Mainnet genesis block initialized.
- [ ] Official nodes recognize the same genesis hash.
- [ ] Independent nodes connect successfully.
- [ ] First valid block produced.
- [ ] First block independently verified.

- [ ] Wallet synchronization verified.
- [ ] First test transaction completed.
- [ ] Explorer data verified.
- [ ] Network health confirmed.

After Launch

- [ ] Monitor block production.
- [ ] Monitor network difficulty.
- [ ] Monitor node synchronization.
- [ ] Monitor peer connectivity.
- [ ] Monitor reorganization events.
- [ ] Monitor transaction processing.
- [ ] Monitor infrastructure resources.
- [ ] Verify supply and reward calculations.
- [ ] Publish launch status.
- [ ] Record and investigate unusual behavior.

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18. MAINNET NO-GO CONDITIONS

Mainnet launch must be delayed if any of the following conditions exist:

1. Unresolved critical or high-severity security vulnerabilities.
2. Consensus rules are incomplete or ambiguous.
3. Independent nodes produce different consensus results.
4. Genesis block hashes do not match.
5. Monetary issuance calculations are incorrect.
6. Unauthorized coin creation is possible.
7. Double-spend protection is not verified.
8. Difficulty adjustment has not been adequately tested.
9. Public testnet has unresolved critical failures.
10. Wallet backup and recovery are unreliable.
11. Official release binaries cannot be verified.
12. Network infrastructure is insufficient for launch.
13. Incident response procedures are not ready.
14. Required legal and regulatory review has not been completed.

A delayed launch is preferable to launching an insecure or inconsistent blockchain.

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19. FINAL APPROVAL

Every approval must be supported by evidence.

Review Area| Responsible Person| Status| Evidence / Report

Protocol Review| TBD| Pending| TBD
Consensus Review| TBD| Pending| TBD
Cryptographic Review| TBD| Pending| TBD
Security Audit| TBD| Pending| TBD
Wallet Testing| TBD| Pending| TBD
Mining Testing| TBD| Pending| TBD
P2P Network Testing| TBD| Pending| TBD
Public Testnet| TBD| Pending| TBD
Release Verification| TBD| Pending| TBD
Infrastructure Review| TBD| Pending| TBD
Legal Review| TBD| Pending| TBD
Final Launch Approval| TBD| Pending| TBD

Launch Decision

- [ ] APPROVED
- [ ] APPROVED WITH DOCUMENTED CONDITIONS
- [ ] REJECTED
- [ ] POSTPONED

Approval Date: _____

Approved By: _____

Final Mainnet Release Version: _____

Genesis Block Hash: _____

Additional Notes:

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20. FINAL PRINCIPLE

SAECOIN mainnet must not launch simply because the software builds or a block can be mined.

A production blockchain requires:

- Correct and deterministic consensus.
- Verified monetary issuance.
- Reliable transaction validation.
- Secure wallet operations.
- Stable peer-to-peer networking.

- Independently reviewed cryptography.
- Reproducible software releases.
- Long-running public testnet results.
- Operational monitoring and incident response.

The mainnet launch decision must be based on verified evidence, not assumptions or deadlines.